

Going down the algorithmic rabbit hole

Approaching affective engagement in montage videos on social media platforms



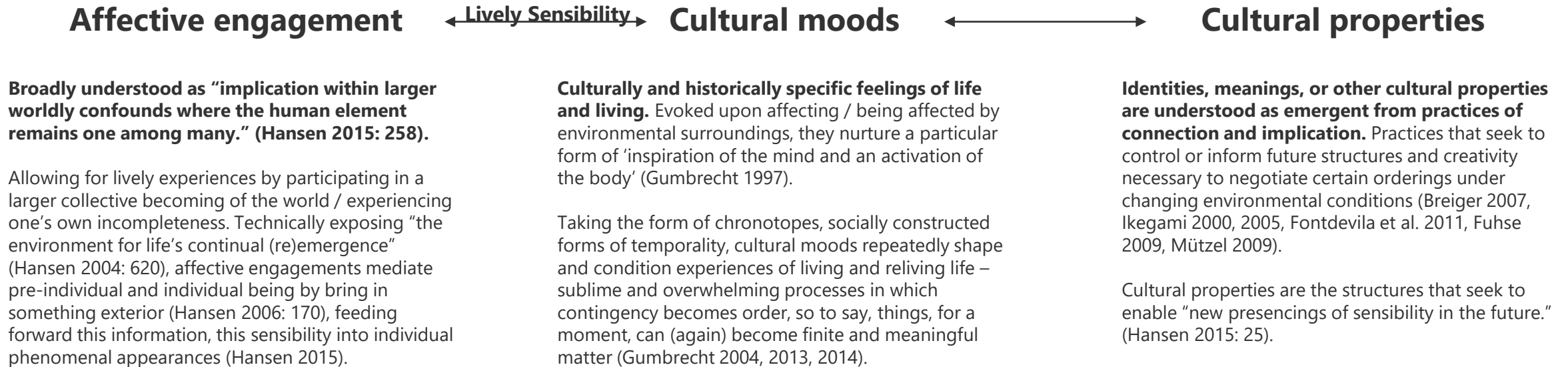
THE UNIVERSITY
of EDINBURGH

Andreas Schellewald, University of Edinburgh
a.o.schellewald@sms.ed.ac.uk

“[I]ncluding the people in a mediated, perhaps mediatized, increasingly datafied age—that’s the task in front of us.”
(Livingstone 2018: 11).

Notes on methodology:

Materialities of communication understood as “the totality of phenomena contributing to the constitution of meaning without being meaning themselves”, analytically “the reconstruction of those processes through which structures of articulated meaning can at all emerge” (Gumbrecht and Pfeiffer 1994: 398).



Video hosting service founded in 2012, acquired by Twitter.

End of service in late 2016, hinging behind innovations of Instagram and Snapchat.



Create, upload, and share six-second long looping videos, so called "Vines".

200M+ active users at the end of 2015.

vines that keep me from ending it all

vines that butter my croissant

vines that i quote daily

vines that cured my cancer

vines that keep me alive



vines that cured my depression clean

vines that are cleaner than your grandma's kitchen

vines that really butter my croissant

vines that toast my buns



Edits

(Source <https://www.youtube.com/watch?v=uoHDSvso8sA>)



Clips

(Source <https://www.youtube.com/watch?v=AfaPJZ0n66k>)



Bits

(Source <https://www.youtube.com/watch?v=ww95zQcAXF0>)



Skits

(Source <https://www.youtube.com/watch?v=TYgd6e4GFnU>)

Montage



YouTube video that is the plain (meaning curated yet neither contextualised nor narrated) sequential display of different six-second long Vines for 10 to 20 minutes.

or

**Moments of life lived, recorded, stored, and (re)distributed.
Put simply, life reassembled in the digital.**

Audio-visual container



**Algorithmically ordered chain of montage (or other related) videos
available on the platform.**

or

**Assemblage in which people, machines, and audio-visual contents
interact and upon which cultural properties (can) emerge.**

Notes on method:

Data gathering:

- YouTube Data v3 API (public access)
- Top 200 seeds for the search query 'vines that' (same results as when using native search via web interface)
- Related videos info for depths 0 and 1 (meaning, get related videos for initial 200 seeds, depth 0, resulting in 3110 new ones, depth 1, for which again get related videos)
- Final network: 48k videos with 169k relations

Network analysis:

Cluster through Louvain community detection algorithm (Blondel et al. 2008)

- Assumption is that YouTube creates relations based on similarity.
- Louvain algorithm generates clusters by optimizing communities towards densely connected nodes.
- Commonly used and reliable community detection method to gain first insight into composition of large networks (Lancichinetti and Fortunato 2011, Papadopoulos et al. 2011, Orman et al. 2012).

Force-directed layout with scaled gravity

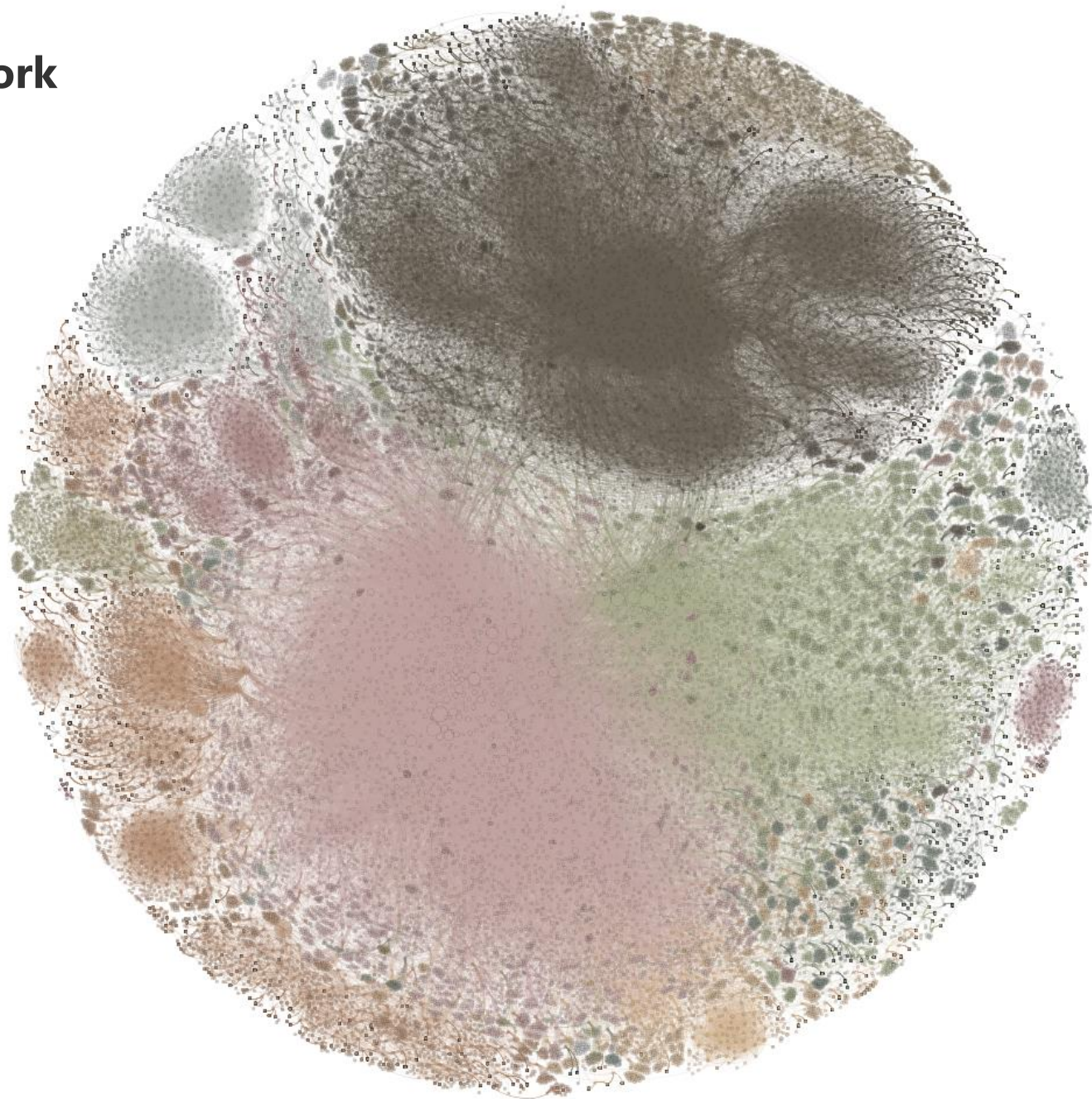
- Central nodes are placed centrally, cluster behave accordingly, gravity scaling enhances this by more compact and comparable spatialization (Bannister et al. 2013).

Content analysis:

Computationally supported text analysis (DiMaggio et al. 2013, Fuhse and Mützel 2011, Mützel 2015, Lewis et al. 2013, Lucas et al. 2015)

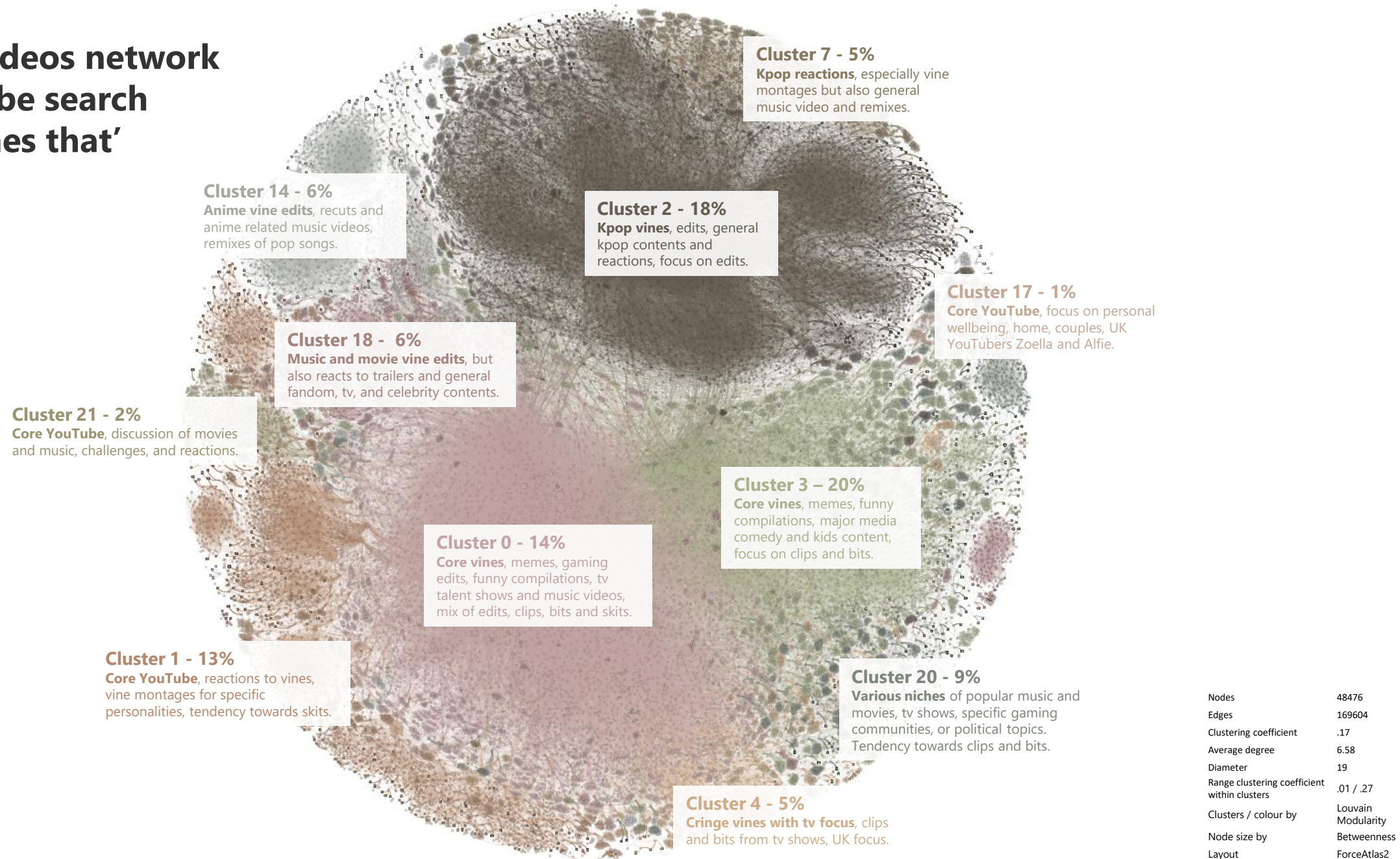
- Assessing themes of clusters through analysis of video titles and video tags (most operable considering size).
- (1) Inductive categorization of lower and upper bound of central (betweenness centrality) and influential nodes (view count and engagement ratio).
- (2) Extraction of central terms through analysis of co-occurrences (Deerwester et al. 1990, Lund and Burgess 1996) by using a graph based ranking model (Mihalcea and Tarau 2004).
- (3) Structural topic modelling for each corpus of video titles with centrality and influence measures as covariates (Roberts et al. 2013)

Related videos network
for YouTube search
query 'vines that'

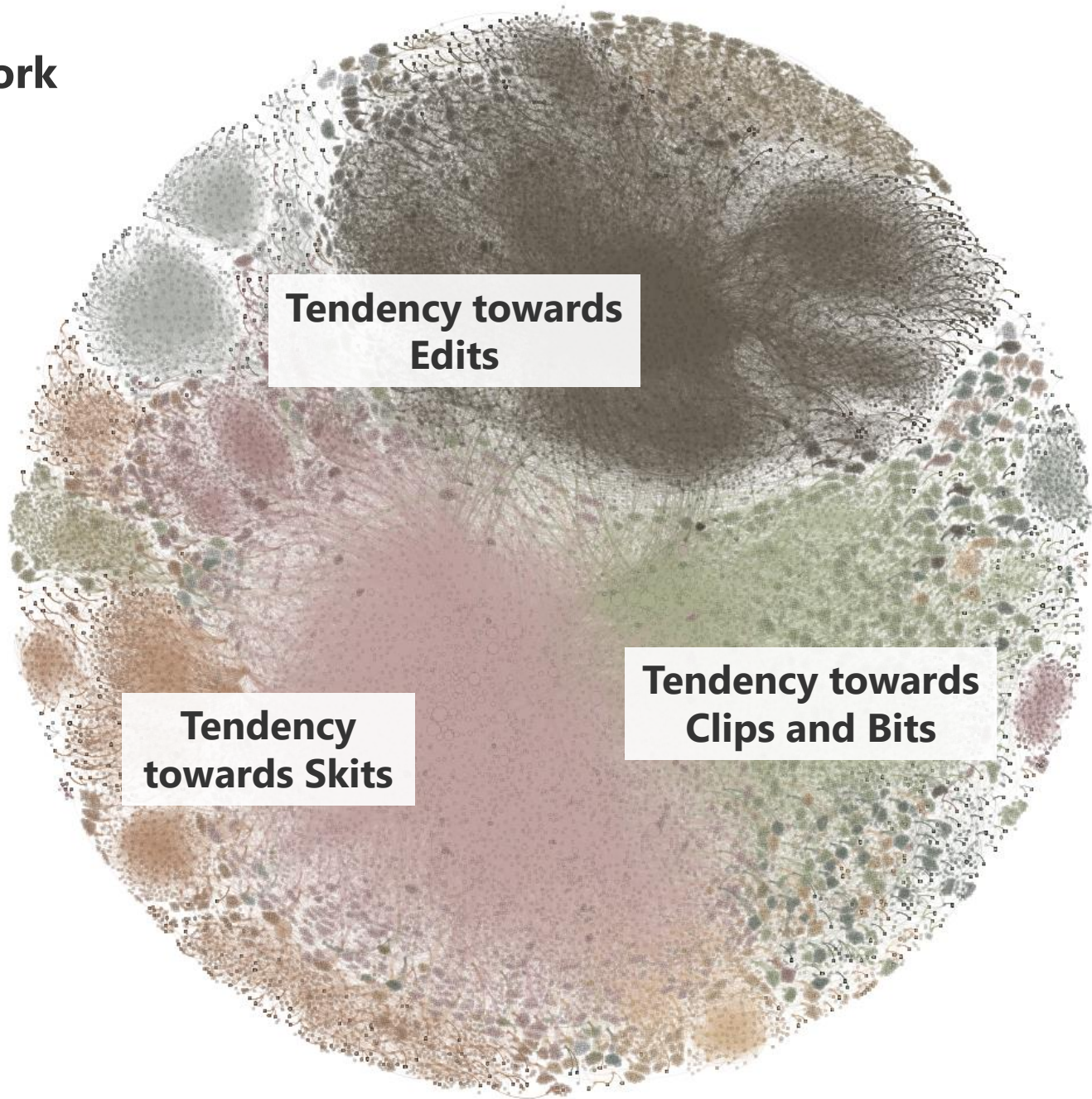


Nodes	48476
Edges	169604
Clustering coefficient	.17
Average degree	6.58
Diameter	19
Range clustering coefficient within clusters	.01 / .27
Clusters / colour by	Louvain
Node size by	Betweenness
Layout	ForceAtlas2

Related videos network for YouTube search query 'vines that'



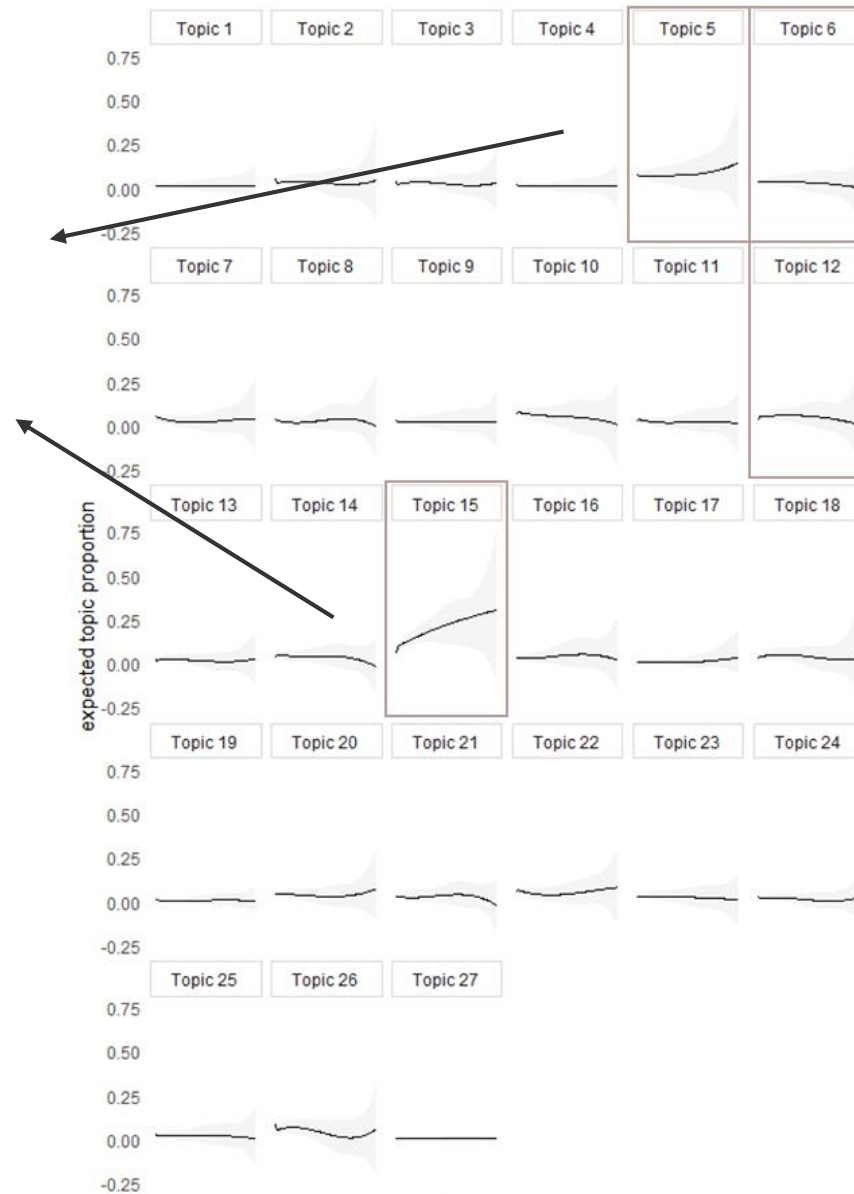
Related videos network
for YouTube search
query 'vines that'



Nodes	48476
Edges	169604
Clustering coefficient	.17
Average degree	6.58
Diameter	19
Range clustering coefficient within clusters	.01 / .27
Clusters / colour by	Louvain
Node size by	Betweenness
Layout	ForceAtlas2

Effects of betweenness centrality on topic distribution for **cluster 0**

Topics that tend to have a higher centrality are core video montages concerning the search query (topics 5 and 15). Most are titled and tagged with 'funny', 'happiness', 'cured me', 'keep me alive', 'laugh'.



Topics 6 and 12 feature similar montages, however have lower centrality. They are titled and tagged with terms like 'depression', 'lost control of life', 'medium rare', 'underrated', 'think about'.

“The target of control is not the production of subjects whose behaviours express internalized social norms; rather, control aims at a never-ending modulation of moods, capacities, affects, and potentialities, assembled in genetic codes, identification numbers, ratings profiles, and preference listings, that is to say, in bodies of data and information.” (Clough 2007: 19).

Thanks!

a.o.schellewald@sms.ed.ac.uk

References

Baecker, D. (2018). 4.0 oder die Lücke die der Rechner lässt. Leipzig: Merve.

Bannister, M.J. et al. (2013). Force-Directed Graph Drawing Using Social Gravity and Scaling. In W. Didimo and M. Patrignani eds., Graph Drawing. GD 2012. Lecture Notes in Computer Science, vol 7704. Berlin: Springer, 414-425.

Blondel, V.D. et al. (2008). Fast unfolding of communities in large networks. In *Journal of Statistical Mechanics*, P10008.

Breiger, R.L. (2007). Cultural Holes: Networks, Meanings, and Formal Practices. Paper presented at the American Sociological Association annual meeting, Culture Section anniversary session, New York.

Clough, P. and Halley, J. eds. (2007). *The Affective Turn: Theorizing the Social*. Durham: Duke University Press.

Deerwester, S.C. et al (1990). Indexing by latent semantic analysis. In *Journal of the American Society for Information Science*, 41(6), 391–407.

DiMaggio, P. et al. (2013). Exploiting affinities between topic modelling and the sociological perspective on culture. In *Poetics*, 41, 570-606.

Fontdevila, J. et al. (2011). Order at the Edge of Chaos: Meanings from Netdom Switchings Across Functional Systems. In *Sociological Theory*, 29(3), 178-198.

Fuhse, J. (2009). The Meaning Structure of Social Networks. In *Sociological Theory*, 27(1), 51-73.

Fuhse, J. and Mützel, S. (2011). Tackling connections, structure, and meaning in networks: quantitative and qualitative methods in sociological network research. In *Qual Quant*, 45, 1067-1089.

Goffman, E. (1974). *Frame Analysis. An Essay on the Organization of Experience*. Boston: Northeastern University Press.

Gumbrecht, H.U. and Pfeiffer, K.L. eds. (1994). *Materialities of Communication*. Stanford: Stanford University Press.

Gumbrecht, H.U. (1997). In 1926. *Living at the Edge of Time*. Cambridge, MA: Harvard University Press.

Gumbrecht, H.U. (2004). *Production of Presence. What Meaning Cannot Convey*. Stanford: Stanford University Press.

Gumbrecht, H.U. (2013). *After 1945. Latency as Origin of the Present*. Stanford: Stanford University Press.

Gumbrecht, H.U. (2014). All that Matters is Invisible: How Latency Dominates our Present. Shulman Lecture Series in Science and the Humanities, April 8, Yale University.

Hansen, M. (2004). The Time of Affect, or Bearing Witness to Life. In *Critical Inquiry*, 20, 584-626.

Hansen, M. (2006). *Bodies in Code. Interfaces with Digital Media*. London: Routledge.

Hansen, M. (2015). *Feed-Forward. On the Future of Twenty-First-Century Media*. Chicago: University of Chicago Press.

Ikegami, E. (2000). A Sociological Theory of Publics. Identity and Culture as Emergent Properties in Networks. In *Social Research*, 67(4), 989-1029.

Ikegami, E. (2005). *Bonds of Civility: Aesthetic Networks and the Political Origins of Japanese Culture*. Cambridge: Cambridge University Press.

Lancichinetti, A. and Fortunato, S. (2011). Limits of modularity maximization in community detection. In *Physical Review*, 84, 066122.

Lewis, S.C. et al. (2013). Content Analysis in an Era of Big Data A Hybrid Approach to Computational and Manual Methods. In *Journal of Broadcasting & Electronic Media*, 57(1), 34-52.

Livingstone, S. (2018). Audiences in an age of datafication: critical questions for media research. In *Television and New Media*, accepted for publication.

Lucas, C. et al. (2015). Computer-Assisted Text Analysis for Comparative Politics. In *Political Analysis*, 25, 254-277.

Lund, K. and Burgess, C. (1996). Producing high-dimensional semantic spaces from lexical co-occurrence. In *Behavior Research Methods, Instruments, and Computers*, 28 (2), 203– 208.

Miheleca, R. and Tarau, P. (2004). TextRank Brining Order into Texts. In *Proceedings of the 2004 Conference on Empirical Methods in Natural Language Processing*.

Mützel, S. (2015). Facing Big Data: Making sociology relevant. In *Big Data & Society*, 1-4.

Mützel, S. (2009). Networks as Culturally Constituted Processes. A Comparison of Relational Sociology and Actor-network Theory. In *Current Sociology*, 57(6), 871-887.

Orman, G.K. et al. (2012). Comparative evaluation of community detection algorithms: a topological approach. In *Journal of Statistical Mechanics*, P08001.

Papadopoulos, S. et al. (2011). Community detection in Social Media. Performance and application considerations. In *Data Mining and Knowledge Discovery*, 24, 515-554.

Ritter, J. (1974). Landschaft. Zur Funktion des Ästhetischen in der modernen Gesellschaft. In J. Ritter, *Subjektivität. Sechs Aufsätze*. Frankfurt a.M.: Suhrkamp, 141-164.

Roberts, M.E. et al. (2013). The structural topic model and applied social science. In *Advances in Neural Information Processing Systems Workshop on Topic Models: Computation, Application, and Evaluation*.

Tsing, A.L. (2015). *The Mushroom at the End of the World. On the Possibility of Life in Capitalist Ruins*. Princeton: Princeton University Press.